Graded Motor-Preparatory Dynamics Preserve Perceptual Decisions over Short Delays

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Abstract

In delayed perceptual reports, the decision is formed before action is permitted, yet response time from the response signal still depends on sensory evidence. This paradox raises the possibility that evidence accumulation leaves a graded trace in motor preparatory circuits rather than ending in a discrete stored choice. We tested this hypothesis in 22 human adults of both sexes (17 women) performing a response-signal random-dot motion task with electroencephalography and electromyography. Participants judged motion direction, withheld the selected response for 3 s, and executed it when the response signal was presented; motion coherence and duration were manipulated. Response time increased when coherence or duration decreased. Fractionating response time at muscle-activation onset showed that these effects were confined to the premotor period. Muscle activations that did not produce overt responses provided a peripheral readout of motor preparation: they were effector-selective, built up during stimulus presentation at a coherence-dependent rate, and continued to ramp during the stimulus-free delay, but no longer at a coherence-dependent rate. These results support an account in which accumulated motion evidence is transmitted to motor preparatory circuits, transformed into a sustained state maintaining the selected response, and shaped by temporal preparation before action is permitted. Electroencephalography revealed a hierarchy: mu–beta-band lateralization tracked motion-evidence accumulation and temporal preparation, whereas the lateralized readiness potential tracked temporal preparation and continued to build up toward the response. Thus, delayed decisions are preserved not as discrete stored choices, but as graded motor-preparatory states that remain dynamically regulated until action is permitted.

Significance statement

Perceptual decisions often cannot be reported immediately, yet the brain must preserve them until action is allowed. By combining brain and muscle recordings in humans, we show that, over short delays, information extracted from a stimulus is not simply converted into a fixed choice and released later. Instead, it remains graded in motor preparatory circuits, where it is maintained and adjusted by expectations about when action can occur. Muscle activations too weak to produce overt responses provided an especially sensitive window onto these hidden preparatory states, offering a novel tool for studying how decisions are maintained and regulated before behavior.

Introduction

Perceptual decision-making refers to the process by which individuals evaluate sensory information to select and execute responses. Most studies investigating the underlying neurocognitive mechanisms in humans have used experimental paradigms in which participants are instructed to respond as soon as they have committed to a choice (Forstmann et al., 2016; O'Connell & Kelly, 2021; Ratcliff & Smith, 2004). This body of work has led to neurally informed models proposing that decisions arise from the accumulation of noisy sensory evidence, which is continuously conveyed to the motor system to support parallel preparation of motor plans (Afacan-Seref et al., 2018; Balsdon et al., 2023; Corbett et al., 2023; Dendauw et al., 2024; Kelly et al., 2021; Selen et al., 2012; Verdonck et al., 2021). However, real-world decisions often involve a temporal gap between commitment and action, and it remains unclear how evidence accumulation models extend to such scenarios.

A canonical paradigm for studying perceptual decision-making is the random-dot motion discrimination task, in which observers judge the direction of motion and indicate their choice via eye or hand movement. Recent studies have introduced a temporal delay between stimulus offset and a response signal (RS) that requires participants to withhold their response until prompted (Achard et al., 2025; Rogge et al., 2022). In this context, one might expect a response to be selected during stimulus presentation, maintained in working memory throughout the delay, and executed when the RS appears. Accordingly, mean response time (RT; measured from RS onset to the response) should be insensitive to stimulus characteristics. Contrary to this prediction, mean RT increases as motion discriminability or duration decrease. This effect cannot be attributed to incomplete decision formation at RS onset, as it persists with delays of up to seven seconds.

The present work aims to uncover the neurocognitive processes through which stimulus information modulates the latency of delayed decision reports. We develop and test a neurally informed theory proposing that sensory evidence is continuously accumulated and transmitted to motor preparatory circuits. Upon reaching an internal threshold, neural activity in motor preparatory circuits transitions into a sustained state that maintains the selected response. This account unifies continuous flow models of perceptual decision-making with electrophysiological findings in humans and nonhuman primates, highlighting shared cortical substrates for evidence accumulation and working memory (Kiani et al., 2008; Rogge et al., 2022; Roitman & Shadlen, 2002; Twomey et al., 2016; Wang, 2008). We further propose that gating inhibition regulates α-motor neuron input from motor preparatory circuits in order to prevent premature responses (Dendauw et al., 2024; Hikosaka, 2007; Mink, 1996). Following RS presentation, sustained motor preparatory activity serves as the starting point for a second phase of accumulation, during which participants sample sensory evidence from the RS to overcome gating inhibition and execute the response, consistent with models of one-choice RT (Ratcliff & Van Dongen, 2011; Smith, 1995).

Building on this extended evidence accumulation theory, our study tests three hypotheses. First, sustained motor preparatory activity may decrease when stimulus discriminability or duration decrease because the quality or quantity of sensory evidence is insufficient for the accumulation process to reach the internal threshold (Ratcliff, 2006). Additional evidence must then be accumulated from the RS to overcome gating inhibition, leading to longer mean RTs (*threshold hypothesis*; Fig. 1A). Alternatively, evidence accumulation trajectories may systematically reach the threshold across all conditions, but sustained motor preparatory activity may continue to increase after threshold crossing as participants temporally prepare for the expected RS (Burle et al., 2010; Heideman et al., 2018). Under a fixed delay interval, reduced stimulus discriminability or duration would leave

less time for temporal preparation to develop before the RS, thereby leading to longer mean RTs (*temporal preparation hypothesis*; Fig. 1B). The third hypothesis represents a combination of these two mechanisms (Fig. 1C). We test these alternative accounts using event-related potentials and spectral electroencephalographic (EEG) measures of effector-selective motor preparation, alongside surface electromyographic (EMG) measures of muscle activation.

Materials and Methods

Participants

Twenty-two participants (17 women; mean age = 20.08; range 18–24) took part in the experiment in exchange for a €20 gift card. The sample size was based on that of our previous study (Achard et al., 2025, Experiment 2), which yielded strong Bayesian evidence (inclusion Bayes factor > 10) for main effects of stimulus discriminability and duration on delayed decision report latencies. Participants were students at Université Marie et Louis Pasteur (France), recruited through online advertisements. They were right-handed, had normal or corrected-to-normal vision, and reported no history of motor, psychiatric, or neurological disorders. They were unaware of the purpose of the experiment and provided written informed consent. The study was approved by the university's research ethics committee (Agreement CERUBFC-2022-01-18-002).

Experimental task

We used an RS variant of the random-dot motion discrimination task, in which participants judged whether a random-dot stimulus moved leftward or rightward, maintained their choice during a 3 s delay, and reported it by pressing a button upon presentation of the

RS at the end of the delay. Stimulus duration (0.3 vs 2 s) and stimulus discriminability were varied factorially. Participants were instructed to press the left button with their left thumb for leftward motion and the right button with their right thumb for rightward motion. Response buttons were housed in a rubberized, ergonomically designed grip manufactured by the Black Box Toolkit Company. The random-dot motion stimulus consisted of white dots (diameter: 0.05°; 71 dots per frame on average) presented within a virtual circular aperture at the center of a black field measuring 26.29°×14.79°. A white noise algorithm was used to control dot motion. In each frame, a percentage *p* of the dots was randomly selected to move in the signal direction (leftward or rightward), while the remaining dots were repositioned randomly. Signal dots moved at a speed of 10.5°/s. Stimulus discriminability was manipulated by varying parameter *p* across three levels (0%, 12% and 50%). This manipulation is commonly referred to as motion coherence. Lower coherence levels reduce stimulus discriminability, making the task more difficult.

Data recording and task performance took place in a dimly lit room. Stimuli were presented on a 34.42 × 19.36 cm LCD monitor (1920 × 1080 resolution; 60 Hz refresh rate). Participants sat in a comfortable chair 75 cm from the monitor, with their hands resting on a support cushion placed on their knees to minimize artifactual motor activity and maximize comfort. They completed 10 blocks of 96 trials each, with self-paced breaks between blocks. Within each block, trials were defined by the factorial combination of motion direction (leftward vs rightward), motion coherence (0% vs 12% vs 50%), and motion duration (0.3 vs 2 s). Each trial type was presented 8 times per block in randomized order, resulting in 160 trials per condition overall. The time course of each trial is illustrated in Figure 2A. A colored fixation cross remained at the center of the screen for the duration of the trial. After a variable jitter (0.6, 0.8, or 1 s, randomly selected on a trial-by-trial basis), the random-dot stimulus was presented, followed by the 3 s delay period. A change in the fixation cross from blue to

yellow served as the RS. Participants were instructed to respond as quickly and accurately as possible upon its presentation. If a response was not made within 4 s of RS onset, the feedback message "Too late, please respond faster" was displayed for 2 s. If a response occurred before the RS, the feedback message "Too early! Please wait for the color change of the fixation cross" was displayed for 2 s. Feedback messages or valid responses were followed by a 1.5 s inter-trial interval, during which a schematic of an eye was displayed. Participants were instructed to maintain fixation on the central cross and avoid blinking until the eye schematic appeared. Trials in which participants failed to respond (0.53%, range 0–5.52%) or responded before the RS (1.59%, range 0.21–5.62%) were excluded from data analyses. Additionally, trials with RTs faster than 0.2 s were classified as anticipations and discarded (0.34%, range 0–2.92%), because such responses are unlikely to reflect the full sequence of RS encoding, the second accumulation phase, and response execution assumed by the model.

EMG acquisition and preprocessing

EMG activity from the *flexor pollicis brevis*, which flexes the thumb at the metacarpophalangeal joint and contributes to flexion at the carpometacarpal joint, was continuously recorded during task performance using two active Ag/AgCl electrodes placed 1 cm apart on the skin over the thenar eminence of each hand. The electrodes were connected to a BioSemi Active Two Mk2 biopotential measurement system (sampling rate = 1024 Hz). After acquisition, the voltage difference between each electrode pair was computed. Left and right bipolar EMG signals were then high-pass filtered at 10 Hz using a third-order Butterworth filter to remove slow drifts unrelated to muscle activity, full-wave rectified, and segmented from -0.3 to 9 s relative to stimulus onset. Muscle activation onsets were detected using the three-step semi-automatic procedure described in Achard et al. (2025). A small

proportion of trials (0.18%, range 0–1.04%) was excluded from data analyses because excessive noise prevented reliable onset detection.

Measurement of muscle activity

Within the extended evidence accumulation theory, the neural drive to muscle fibers directly reflects motor preparatory activity that exceeds the level of gating inhibition. This architecture motivates the analysis of several variables derived from EMG signals. First, RT in each trial was fractionated based on the onset of muscle activation leading up to the response, yielding two latencies: the pre-motor execution time (PMET) and the motor execution time (MET; Fig. 2C). Both PMET and MET are determined by the accumulation of evidence sampled from the RS, but PMET additionally depends on the difference between the starting point of this second accumulation phase and the level of gating inhibition (Fig. 1 and Fig. 2B,C). Because each of the three hypotheses predicts a lower starting point of the second accumulation phase when motion coherence or duration decreases, they all predict corresponding main effects on mean PMET, with longer mean PMETs in lower-coherence and shorter-duration conditions. By contrast, mean MET should not be modulated by motion coherence or duration, because the time required to execute the response once muscle activation has begun does not depend on this starting point. Moreover, the temporal preparation hypothesis does not predict an interaction between motion coherence and duration on mean PMET, because coherence-related differences in sustained motor preparatory activity, and therefore in the starting point of the second accumulation phase, should not vary with motion duration (Fig. 1B, vertical black arrows). By contrast, the threshold and combined hypotheses allow such an interaction, because both the quality and the quantity of motion evidence determine whether motor preparatory activity reaches the internal threshold. Under these hypotheses, increasing stimulus duration may have two consequences. If the additional motion evidence available in the 2 s condition allows trajectories to reach the

internal threshold across coherence levels, coherence-related differences in sustained motor preparatory activity, and therefore in the starting point of the second accumulation phase, may be reduced relative to the 0.3 s condition, yielding a smaller coherence effect on mean PMET. Conversely, if the additional motion evidence mainly benefits high-coherence trials, such that low-coherence trajectories reach the internal threshold less often, coherence-related differences in sustained motor preparatory activity and starting point may be larger in the 2 s than in the 0.3 s condition, yielding a larger coherence effect on mean PMET (Fig. 1A and 1C, vertical black arrows). Thus, unlike the temporal preparation hypothesis, the threshold and combined hypotheses can predict a motion coherence × duration interaction on mean PMET, but the direction of this interaction depends on how the additional motion evidence provided by the 2 s duration changes the relative probability of reaching the internal threshold across coherence levels.

Second, previous work has demonstrated the presence of muscle activations that do not culminate in an overt response in both the standard and RS versions of the random-dot motion task (Achard et al., 2025; Servant et al., 2021). We refer to these events as partial muscle activations (PMAs). The extended evidence accumulation theory predicts that PMAs occur at the single-trial level when noisy motor preparatory activity temporarily exceeds gating inhibition (Fig. 2B,C), without generating sufficient muscle activation to produce an overt response. Although PMAs may occur at any point during a trial, their instantaneous rate should depend on the distance between the mean level of motor preparatory activity and the level of gating inhibition. To approximate this time-varying rate while maintaining a reasonable signal-to-noise ratio, we computed the average number of PMAs per 0.1 s time bin, separately for each subject, EMG channel, and condition. The PMA rate should approach zero in the EMG channel corresponding to the non-selected response. By contrast, in the channel associated with the selected response, it should provide an indirect proxy for the

mean dynamics of motor preparatory activity illustrated in Figure 1, at least between stimulus onset and RS onset. Following the RS, the strong drive produced by the second phase of evidence accumulation should reduce the PMA rate, because motor preparatory activity that exceeds gating inhibition will predominantly lead to an overt response.

EEG acquisition and preprocessing

EEG signals were recorded continuously during task performance from 64 active Ag/AgCl electrodes positioned according to the international 10–20 system, using the same BioSemi ActiveTwo system and sampling rate as the EMG recordings. Vertical and horizontal eye movements were recorded using electro-oculogram (EOG) electrodes: two placed above and below the right eye to measure vertical movements, and two placed at the outer canthi of the eyes to measure horizontal movements. The data were preprocessed using custom scripts together with functions from the MNE-Python toolbox (v1.9.0). Horizontal and vertical EOG signals were derived by computing the voltage difference between the corresponding electrode pairs. EEG channels exhibiting excessive variance or drifts were interpolated using the spherical spline method, and the data were re-referenced to the average of all electrodes. Ocular artifacts were corrected using independent component analysis (ICA). For ICA decomposition only, the continuous EEG data were high-pass filtered at 1 Hz to remove slow drifts and low-frequency noise that could impair artifact isolation, and then epoched from −0.6 to 9 s relative to stimulus onset. ICA was implemented using the default parameters of MNE's ICA function. Components associated with eye movements were identified using MNE's *find_bads_eog* function, which detects EOG-related components based on a z-score-transformed Pearson correlation between ICA components and EOG signals filtered in the 1–10 Hz range. Components with z-scores exceeding a threshold of 3 were masked at each iteration, and z-scores were recalculated on the remaining components. This iterative procedure enables adaptive identification of ocular artifacts and, together with

196 visual inspection, ensured robust detection of eye-movement-related components. Up to four
197 components were removed per participant. The resulting ICA weights were then applied to
198 EEG data that had been filtered between 0.1 and 40 Hz before segmentation. All EEG and
199 EOG preprocessing filters were finite impulse response filters with a Hamming window. The
200 epoched data were baseline-corrected using the 0.1 s window preceding stimulus onset, and
201 epochs were excluded from further analysis if any scalp channel exceeded an absolute
202 amplitude of 150 μV (M = 4.65%, range 0–21.98%). Finally, the EEG data were transformed
203 to current source density (CSD) using the corresponding MNE function with default
204 parameters, to reduce the spatial blurring effects of volume conduction (e.g., Kelly et al.,
205 2021; Kelly & O'Connell, 2013; Twomey et al., 2016). After applying EEG, EMG, and
206 behavioral trial exclusion criteria, an average of 85.52% of trials remained for analysis (range
207 48.4–98.33%, SD = 11.35).

208 **Measurement of motor preparation EEG signals**

209 EEG studies have identified two main markers of motor preparation: the lateralized
210 readiness potential (LRP), a time-domain event-related potential, and mu–beta-band spectral
211 amplitude lateralization (MBL), a time-frequency measure capturing asymmetric suppression
212 of mu–beta oscillations over motor cortices. Both markers are computed as difference
213 waveforms by contrasting activity over motor areas contralateral and ipsilateral to the
214 responding effector. The LRP is a negative-going potential that precedes unilateral
215 movements and is commonly interpreted as reflecting effector-specific motor activation in
216 contralateral motor-related regions, indexing late-stage response preparation (Gratton et al.,
217 1988; de Jong et al., 1988; Eimer, 1998). MBL reflects the relative decrease in mu–beta
218 amplitude (8–30 Hz) over the contralateral hemisphere as motor plans are formed, indexing
219 the lateralized preparatory modulation of motor cortical activity (Pfurtscheller & Lopes da
220 Silva, 1999). In standard perceptual decision-making tasks, both signals have been shown to

track the process of evidence accumulation: their buildup rate scales with the strength of sensory evidence, and both reach a stereotyped amplitude at response execution (Kelly et al., 2021; Kelly & O'Connell, 2013; Rogge et al., 2022; van Vugt et al., 2014). Consequently, the two signals are often used interchangeably in the literature, including in neurally informed modeling studies (Afacan-Seref et al., 2018; Corbett et al., 2023; Kelly et al., 2021; Murphy et al., 2016), with the LRP typically preferred when high temporal resolution is required. However, recent empirical evidence has challenged this assumption of functional equivalence between the LRP and MBL, as discussed below.

To our knowledge, only two studies have examined LRP and/or MBL dynamics in RS variants of the random-dot motion task (Rogge et al., 2022; Twomey et al., 2016). In the variant used by Twomey et al. (2016), the random-dot motion stimulus was presented for 1.4 s and was immediately followed by a letter cue serving as the RS, with no delay between the two events. Following stimulus onset, MBL increased and then transitioned into a sustained state, presumably reflecting short-term maintenance of the selected response, consistent with the extended evidence accumulation theory. However, MBL did not exhibit a coherence-dependent buildup rate, possibly because the difference in motion coherence between conditions was relatively small (20% vs 30%). Possible influences of coherence and temporal preparation on sustained MBL activity were not assessed. The motor-related nature of MBL was confirmed in a control condition in which the stimulus–response mapping was delivered by the RS: in this case, MBL remained silent until RS onset. In another study, Rogge et al. (2022) presented the random-dot motion stimulus for 1.2 s and introduced a 0.2 s delay between stimulus offset and RS onset, implemented as a change in fixation from a point to a cross. The range of motion coherence levels was much larger than in Twomey et al. (2016). Both LRP and MBL were analyzed, with MBL further divided into alpha power lateralization (8–12 Hz) and beta power lateralization (13–30 Hz). The alpha and beta components showed

temporal dynamics comparable to those observed by Twomey et al. (2016), although their buildup rate decreased as motion coherence decreased. The effects of coherence and the potential impact of temporal preparation on sustained alpha and beta power lateralization were not examined. Importantly, the LRP remained silent until RS onset, although a transient negative deflection was observed approximately 300 ms after motion onset in the highest coherence condition. Following the RS, a negative deflection emerged, with a buildup rate that was not modulated by coherence, peaking just before the response. Together, these findings suggest a functional dissociation between MBL and LRP within the motor preparation hierarchy. Specifically, the processes indexed by the LRP appear to reflect the final stage of motor preparation, whereas those indexed by MBL appear to occur upstream, contributing to the motor-level representation of evidence accumulation and the short-term maintenance of the selected response.

In line with this dissociation, we hypothesized that motor preparation trajectories predicted by the extended evidence accumulation theory would manifest in MBL before RS onset, and in the LRP thereafter. Together with the EMG-derived measures described above, this mapping provides a complementary framework for evaluating the three hypotheses illustrated in Figure 1. To measure MBL, we extracted MB amplitude (8–30 Hz) over time for each electrode by applying a short-time fast Fourier transform to the CSD signals from each epoch, using a 0.3 s window and a 0.1 s step. To identify electrode sites exhibiting maximal effector-selective activity, we computed a grand-average topography for the -0.2 to 0 s interval relative to the response by subtracting, at each electrode site, the mean MB amplitude for right-hand trials from that for left-hand trials, irrespective of experimental condition. An analogous topography was constructed using time-domain CSD amplitudes. The resulting maps revealed overlapping extrema at electrodes C3 and C4 (Fig. 4C,D), which were therefore selected for subsequent analyses. MBL and LRP were computed by subtracting, for

each trial, the amplitude at the electrode ipsilateral to the responding hand from that at the contralateral electrode, using MB and CSD signals, respectively.

Statistical Analysis

Behavioral, EMG, and EEG data were analyzed using repeated-measures ANOVAs, with the significance threshold set at $\alpha = .05$. The sphericity assumption was evaluated using Mauchly's test, and violations were corrected with the Greenhouse–Geisser procedure. To maximize statistical power, analyses included both correct and incorrect trials; analyses restricted to correct trials yielded comparable results. ANOVAs on response accuracy, mean RT, mean PMET, and mean MET included motion coherence and motion duration as factors. As described above, the instantaneous PMA rate was approximated by computing the average number of PMAs per 0.1 s time bin. Accordingly, ANOVAs on PMA rate included an additional time-bin factor and were conducted separately across four periods of interest, allowing direct assessment of the predictions illustrated in Figure 1. Each period was time-locked to one of the four main events within a trial: stimulus onset, stimulus offset, RS onset, and the response. The stimulus-onset-locked analysis covered the 2 s following stimulus onset (20 time bins). The stimulus-offset-locked analysis covered the interval from 1 to 3 s after stimulus offset (20 time bins), to avoid redundant inclusion of motion-evidence accumulation in the 0.3 s duration condition. The RS-onset-locked analysis covered the 0.2 s following RS onset (2 time bins), ensuring equal trial counts across bins. Finally, the response-locked analysis focused on the 0.2 s preceding the button press (2 time bins). In what follows, we refer to these four analysis windows as the *stimulus*, *delay*, *RS*, and *response periods*. LRP and MBL were analyzed using the same strategy, except that the dependent variable in each bin was mean signal amplitude.

Results

For clarity and conciseness, we summarize the ANOVA results for accuracy, mean RT, mean PMET, and mean MET in Table 1 (significant effects only) and Table S1 (full statistical results, including effect sizes). The ANOVA results for PMAs, MBL, and LRP are reported in Table 2 and Tables S2–S4, using the same organizational structure. We present the results in reverse processing order, proceeding from behavior to EMG and then to EEG.

Behavior

Figures 3A and 3B show response accuracy and mean RT, respectively, as a function of motion coherence and duration. Accuracy decreased as motion coherence decreased and was higher in the 2 s condition than in the 0.3 s condition, but this duration benefit was observed only at the intermediate 12% coherence level, yielding a significant motion coherence × duration interaction. Mean RT increased as motion coherence and duration decreased, with the effect of coherence more pronounced in the 0.3 s condition, also yielding a significant motion coherence × duration interaction.

Motion coherence and duration both modulate mean PMET but not mean MET

Mean PMET and mean MET for each experimental condition are shown in Figures 3C and 3D, respectively. Mean PMET exhibited the same pattern of statistical effects as mean RT. In contrast, mean MET was not significantly affected by any experimental factor. These findings are consistent with the general predictions of the extended evidence accumulation theory. The motion coherence × duration interaction observed for mean PMET reflected a larger motion coherence effect in the 0.3 s condition than in the 2 s condition. This pattern suggests that coherence-related differences in the starting point of the second accumulation phase were smaller in the 2 s condition than in the 0.3 s condition.

PMA rate indexes motion evidence accumulation and temporal preparation

For each period of interest, Figure 4A shows the average number of PMAs per 0.1 s time bin in the response-associated EMG channel, whereas Figure 4B shows the corresponding data for the opposite EMG channel. As expected, PMA rates in the opposite EMG channel were near zero across conditions and periods. Consequently, subsequent analyses were restricted to the response-associated EMG channel. An ANOVA on the stimulus period data (0–2 s window relative to stimulus onset) revealed a significant time bin × motion coherence interaction. PMA rate increased over time, and the slope of this ramping activity became shallower as motion coherence decreased, consistent with an evidence accumulation mechanism. The analysis also showed a significant time bin × duration interaction, reflecting steeper ramping in the 0.3 s than in the 2 s duration condition. This effect is unlikely to reflect greater allocation of attentional resources in the 0.3 s condition, because stimulus duration was unpredictable at trial onset. Instead, it is more consistent with the combined hypothesis: temporal preparation begins earlier in the 0.3 s than in the 2 s condition, and this earlier onset accounts for the observed interaction if temporal preparation increases more steeply than coherence-averaged evidence accumulation.

Figure 4A shows that PMA rate continues to increase during the delay period. This increase is best attributed to temporal preparation, because the ANOVA on the corresponding data (1–3 s window relative to stimulus offset) indicated that the time bin × motion coherence interaction, used here as a diagnostic marker of evidence accumulation, was no longer significant and did not interact with duration. By contrast, the main effect of time bin remained significant, indicating continued ramping. Thus, the threshold hypothesis alone cannot account for these data. The analysis also revealed a significant time bin × duration interaction. PMA rate was generally higher in the 2 s than in the 0.3 s condition, and this difference increased over time. Although this interaction is opposite in direction to that

observed during the stimulus period, both effects likely arise from the same underlying mechanism. The apparent reversal follows from the different temporal alignment: at stimulus offset, evidence accumulation has only just started in the 0.3 s condition, whereas in the 2 s condition it has likely already reached the internal threshold and transitioned into temporal preparation. Finally, we observed a significant duration × motion coherence interaction. PMA rate decreased with lower motion coherence, and this effect was more pronounced in the 2 s condition. At first glance, this pattern seems difficult to reconcile with the mean PMET results. Whereas mean PMET suggests larger coherence-related differences in the starting point of the second accumulation phase in the 0.3 s condition, the PMA results suggest larger coherence-related differences in sustained motor preparatory activity in the 2 s condition. One possible explanation is that, in the 2 s condition, motor preparation occasionally became high enough around RS onset to produce anticipatory responses, especially at high motion coherence. Because these trials were excluded from the main analyses, their exclusion may have selectively removed some of the most strongly prepared trials in the high motion coherence condition, thereby reducing coherence-related differences near RS onset and in the starting point of the second accumulation phase estimated from valid-response trials. This interpretation is consistent with the numerical trend toward more anticipatory responses in the 2 s condition, particularly at high motion coherence (Figure S1), but is unlikely to fully explain the reversal because such responses were rare. A complementary possibility is that temporal preparation was strategically reduced as motor preparatory activity approached gating inhibition, compressing differences in PMA rate among the most strongly prepared trials. This possibility is consistent with the apparent PMA-rate trajectory leveling off shortly before RS onset in the 2 s high motion coherence condition (Figure 4A).

As predicted, the PMA rate decreased after RS onset, because the strong drive produced by the second phase of accumulation reduces the probability that motor preparatory

activity exceeding gating inhibition remains partial rather than culminating in an overt response. Accordingly, the ANOVAs conducted on both the RS period (0–0.2 s relative to RS onset) and the response period (-0.2–0 s relative to the response) showed a main effect of time bin. There was also a significant time bin × duration interaction, reflecting a steeper decrease in PMA rate in the 2 s than in the 0.3 s condition. This pattern is largely a consequence of the higher PMA rate at RS onset in the 2 s condition. Coherence-related effects were not significant despite the numerical trend noted in the previous paragraph, possibly due to limited power given that only two bins were considered.

LRP tracks temporal preparation during the delay and the final phase of motor preparation preceding the response

Figure 4C shows the LRP for each condition and period. Although the LRP showed a slight numerical trend suggestive of evidence accumulation during the stimulus period, the ANOVA did not reveal any significant effect. During the delay, however, the analysis showed a clear effect of time bin: the LRP became progressively more negative, consistent with temporal preparation for the upcoming response. This coherence-independent ramping activity persisted after the RS and reached its largest negative amplitude at the time of the response. These findings are partially consistent with our hypothesis. Although the LRP primarily reflects the final stage of motor preparation, it also indexes temporal preparation during the delay for the selected response, suggesting that the underlying neural circuits are informed about the outcome of the motion evidence accumulation process.

MBL tracks motion evidence accumulation and temporal preparation

Figure 4D shows the MBL amplitude for each condition and period. The ANOVA conducted on the stimulus-period data revealed a significant interaction between time bin and motion coherence. MBL exhibited ramping activity whose buildup rate scaled with motion

coherence, consistent with an evidence accumulation mechanism. Unlike the PMA rate, MBL amplitude showed a slight decrease before transitioning to sustained activity, at least in the highest coherence conditions. During the delay, sustained MBL activity increased over time (main effect of time bin), suggesting that MBL also reflects temporal preparation. In contrast to the PMA rate, none of the effects of motion coherence or duration reached statistical significance. After RS onset, MBL amplitude did not show any clear modulation, aside from an attenuation before the response that was independent of the experimental factors. We do not interpret this effect further, as the temporal resolution of MBL is too low to support reliable inferences during the RS and response periods.

Discussion

A central question for evidence accumulation models is how a decision is preserved when commitment and action are separated in time. A simple possibility is that the choice formed during stimulus presentation is stored throughout the delay and released when action is permitted. The present findings argue against this passive storage-and-release view. Replicating previous work (Achard et al., 2025; Rogge et al., 2022), mean RT increased as motion coherence or duration decreased, even though RT was measured from RS onset to the response, 3 s after stimulus offset. Thus, delayed report latency remained tied to the quality and quantity of motion evidence available before the delay.

The key contribution of the present study is to identify the motor-preparatory mechanisms underlying this behavioral effect. EMG was particularly informative because it provided single-trial access to two complementary events: PMAs, whose rate served as a time-resolved proxy of motor preparatory activity, and the onset of response-related muscle activation, which allowed RT to be fractionated into PMET and MET (Coles et al., 1985;

Servant et al., 2021). The PMA results first supported the general architecture of the extended evidence accumulation theory. PMAs were nearly absent in the opposite EMG channel but clearly expressed in the response-associated channel, indicating that they reflected effector-selective motor preparation rather than nonspecific muscular noise. In this channel, PMA rate increased during stimulus presentation with a buildup rate that scaled with motion coherence, consistent with the accumulation of motion evidence at the motor-preparatory level (Donner et al., 2009; Kelly & O'Connell, 2013; Rogge et al., 2022; Selen et al., 2012). PMA rate then remained elevated throughout the delay, indicating that the selected response was maintained as a sustained motor preparatory state. Finally, PMA rate decreased after RS onset, consistent with the idea that processing of the RS triggers a second accumulation phase that adds drive to this sustained preparatory state, making motor activity more likely to culminate in overt muscle activation rather than remain partial. Together, these observations support the core proposal that sensory evidence is accumulated and transmitted to motor preparatory circuits, where it can be maintained until action is permitted.

RT fractionation then linked this preparatory state to delayed report latency and helped evaluate the three hypotheses laid out in the Introduction to explain why motion coherence and duration modulate mean RT. Within the extended evidence accumulation theory, PMET is sensitive to the starting point of the second accumulation phase relative to gating inhibition, whereas MET reflects the time required to execute the response once muscle activation has begun. The finding that motion coherence and duration modulated mean PMET, but not mean MET, localizes the behavioral effects to processes preceding muscle activation onset and indicates that reductions in motion coherence or duration affected the starting point of the second accumulation phase.

The critical question is therefore why this starting point varied across conditions. The answer is most consistent with the combined hypothesis: the starting point of the second

accumulation phase reflected both the evidence-dependent state inherited from motion processing and a time-dependent preparatory process developing during the delay. PMA dynamics during the delay period provide the first constraint. PMA rate ramped throughout the delay period, when the stimulus was off, but without the coherence-dependent buildup rate observed during stimulus presentation. This pattern is difficult to reconcile with the threshold hypothesis alone, in which delay-period activity should primarily reflect the state reached at the end of motion processing. Instead, it is consistent with temporal preparation for the expected RS, in line with work showing that temporal expectations shape anticipatory motor activity (Burle et al., 2010; Heideman et al., 2018; Ramayya et al., 2025). At the same time, the temporal preparation hypothesis alone is also insufficient, because motion duration affected decision accuracy, not only delayed report latency. Accuracy was higher in the 2 s than in the 0.3 s condition at intermediate motion coherence, where floor and ceiling effects were less limiting. Thus, longer stimulus presentations improved the quality of the decision state inherited from motion processing, not merely the time available for temporal preparation before the RS. The motion coherence × duration interactions observed for both mean PMET and PMA rate during the delay provide convergent evidence for this conclusion. These interactions are not expected under the temporal preparation hypothesis alone, which predicts that coherence-related differences in the starting point of the second accumulation phase should not depend on motion duration.

The motion coherence × duration interaction on PMA rate during the delay, with larger coherence-related differences in the 2 s than in the 0.3 s condition, further specifies how motion duration shaped the preparatory state inherited from the first accumulation phase. This direction is compatible with the threshold and combined hypotheses if the additional motion evidence available in the 2 s condition primarily benefited high-coherence trials. In the 0.3 s condition, accumulation was curtailed early, leaving limited time for high- and low-

coherence trajectories to differentiate before the delay. In the 2 s condition, by contrast, high-coherence trajectories were more likely to reach the internal threshold and enter a high sustained preparatory state, whereas lower-coherence trajectories were more likely to remain below threshold. Thus, increasing motion duration appears to have amplified coherence-related differences in sustained motor preparatory activity, yielding larger coherence-related differences in delay-period PMA rate. Although an unbounded accumulation account could also predict this pattern if participants accumulated all available motion evidence, prior work speaks against this interpretation. Random-dot decisions with imposed stimulus durations, including delayed-response conditions, can be described by an accumulation-to-threshold mechanism (Mazurek et al., 2003), and monkeys can commit to a choice before the end of the motion stimulus, reducing the impact of later motion evidence (Kiani et al., 2008). The larger coherence-related differences in delay-period PMA rate are therefore more consistent with accumulation to an internal threshold than with unrestricted accumulation of motion evidence throughout the stimulus.

The EEG results provide complementary evidence about how these dynamics unfold across the cortical motor-preparation hierarchy. MBL tracked motion-evidence accumulation during stimulus presentation, with a buildup rate that scaled with motion coherence, and then remained sustained during the delay while continuing to increase over time. This profile is consistent with the view that alpha- and beta-band lateralization carry decision-related information into motor cortical circuits before final response implementation (Rogge et al., 2022; Twomey et al., 2016). The present findings refine this view by showing that the sustained MBL state is not merely static maintenance, but is also dynamically shaped by temporal preparation for the expected RS. The LRP showed a different profile: it did not reliably track motion-evidence accumulation, but ramped during the delay, continued after RS onset, and reached its largest amplitude near the response. Thus, temporal preparation was not

confined to upstream motor representations, but also shaped later-stage effector-selective motor preparation.

This hierarchical pattern helps explain why PMA rate was the most integrative and sensitive readout of motor preparation in the present study. MBL captured evidence-dependent buildup and delay-period temporal preparation, whereas the LRP captured temporal preparation and later-stage motor preparation. However, neither EEG measure revealed the full coherence- and duration-dependent structure of sustained activity during the delay. PMA rate did so, while also capturing stimulus-period buildup, sustained maintenance, and the post-RS transition from partial activation toward overt execution. This broader sensitivity likely reflects the fact that PMAs are single-trial events measured directly from muscle activity, where signal-to-noise ratio is higher than for scalp EEG markers. In a continuous-flow architecture, upstream and later-stage motor-preparatory processes should converge onto the net motor drive transmitted toward the muscles. PMA rate therefore provides a precise peripheral readout of how hierarchical motor-preparatory processes are expressed throughout the decision-to-action sequence, making it a promising tool for studying how decision evidence, temporal expectation, urgency, inhibitory control, and action constraints jointly shape motor preparation (Balsdon et al., 2023; Dendauw et al., 2024; Ramayya et al., 2025; Rogge et al., 2022; Servant et al., 2021). A natural next step is to use this sensitivity to constrain formal models that estimate threshold-reaching probabilities and the contribution of temporal preparation to the sustained preparatory state.

In conclusion, delayed decision reports are not generated by storing an abstract completed choice and releasing it on demand. Instead, prior sensory evidence remains embodied in motor preparatory circuits, where it is maintained, reshaped by temporal expectations, and converted into overt action when the appropriate signal appears. By combining RT fractionation, PMA dynamics, and EEG markers of cortical motor preparation,

the present study shows how evidence accumulation can bridge perception, decision maintenance, and action. This extends evidence accumulation models beyond immediate reports by showing that continuous flow into the motor system preserves decisions over time, reframing motor preparation as an active substrate of decision maintenance.

Figures

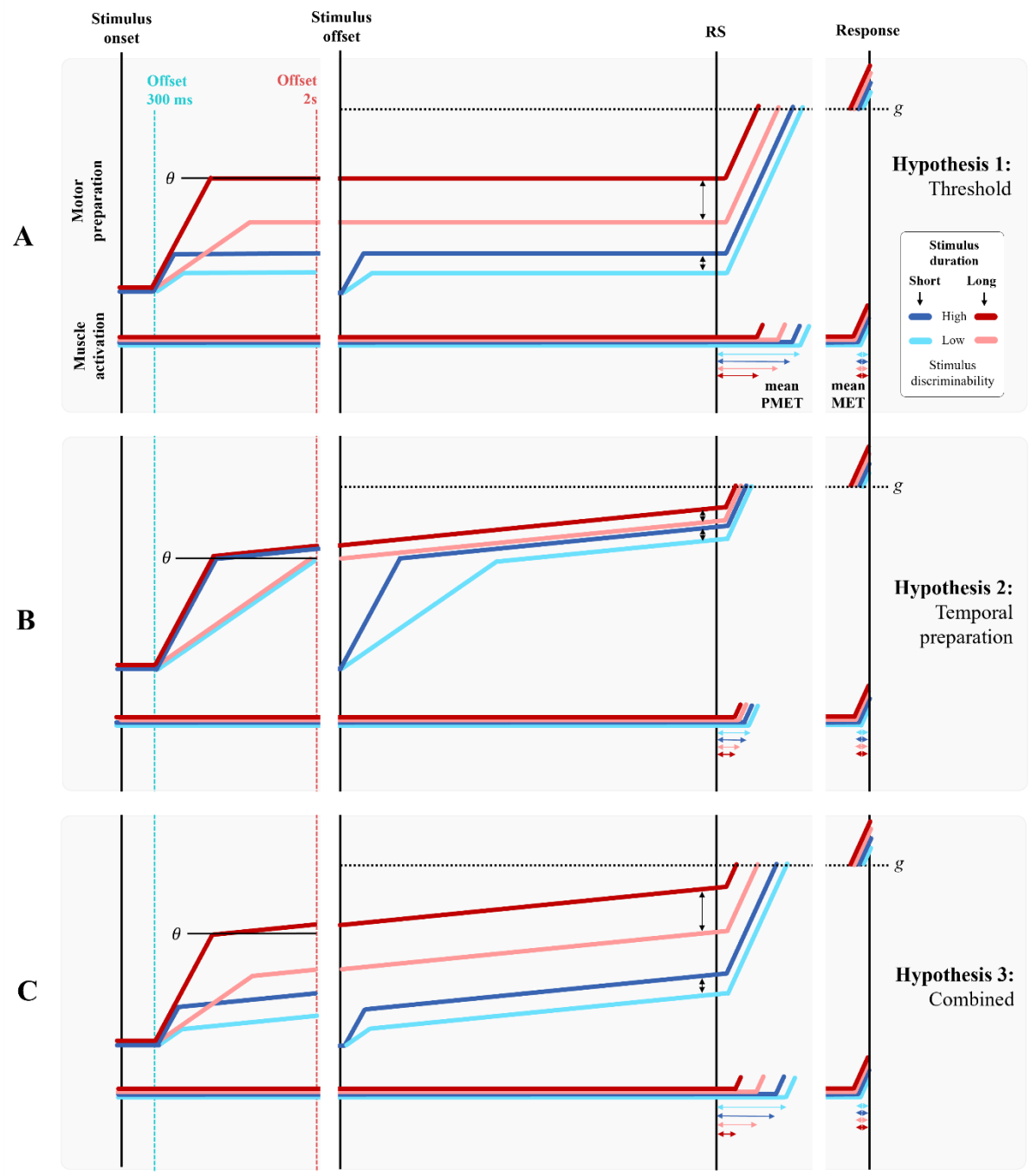


Fig 1. Extended evidence accumulation theory and predictions. The extended evidence accumulation theory delineates the neurocognitive mechanisms underlying perceptual decision reporting with short-term delays. During stimulus presentation, sensory evidence is accumulated and relayed to motor preparatory circuits. Once motor preparatory activity reaches an internal threshold, denoted by parameter θ, it transitions into a sustained state that maintains the selected response throughout the delay period. During this interval, gating inhibition (parameter g) prevents premature activation of α-motor neurons innervating muscle fibers. Processing of the response signal (RS) provides the additional drive required to overcome gating inhibition and execute the response.

Specifically, the theory posits a second phase of evidence accumulation in which evidence is sampled from the RS and transmitted to motor preparatory circuits, where it sums with the sustained state. Muscle activation is initiated when motor preparatory activity exceeds gating inhibition, and the response is executed when the necessary force has been generated. Building on this framework, the present study tests three alternative hypotheses accounting for previous reports that the mean latency of delayed decision reports increases following reductions in stimulus discriminability or duration. Panels A–C illustrate the expected mean trajectories of motor preparation and muscle activation under each hypothesis, across four experimental conditions defined by the factorial combination of stimulus discriminability (low vs high) and stimulus duration (short vs long). Trajectories are shown time-locked to the four events within each trial: stimulus onset, stimulus offset, RS onset, and the response. Vertical black arrows indicate, for each duration condition, the predicted coherence-related differences in sustained motor preparatory activity during the delay. Following RS-related sensory encoding, these differences are expected to determine the relative starting levels of the second accumulation phase. Predictions at the level of motor preparation will be tested using electroencephalography, whereas predictions at the level of muscle activation will be tested using electromyography. Each hypothesis predicts an increase in mean response time (RT; from RS onset to the response) following reductions in stimulus discriminability or duration. This effect should be evident only in mean pre-motor execution time (PMET), corresponding to the latency between RS onset and muscle activation onset, whereas mean motor execution time (MET; from muscle activation onset to the response) should remain unchanged. Colored horizontal arrows indicate the predicted mean PMET and MET for each condition, using the same color code as the trajectories. See text for details of each hypothesis.

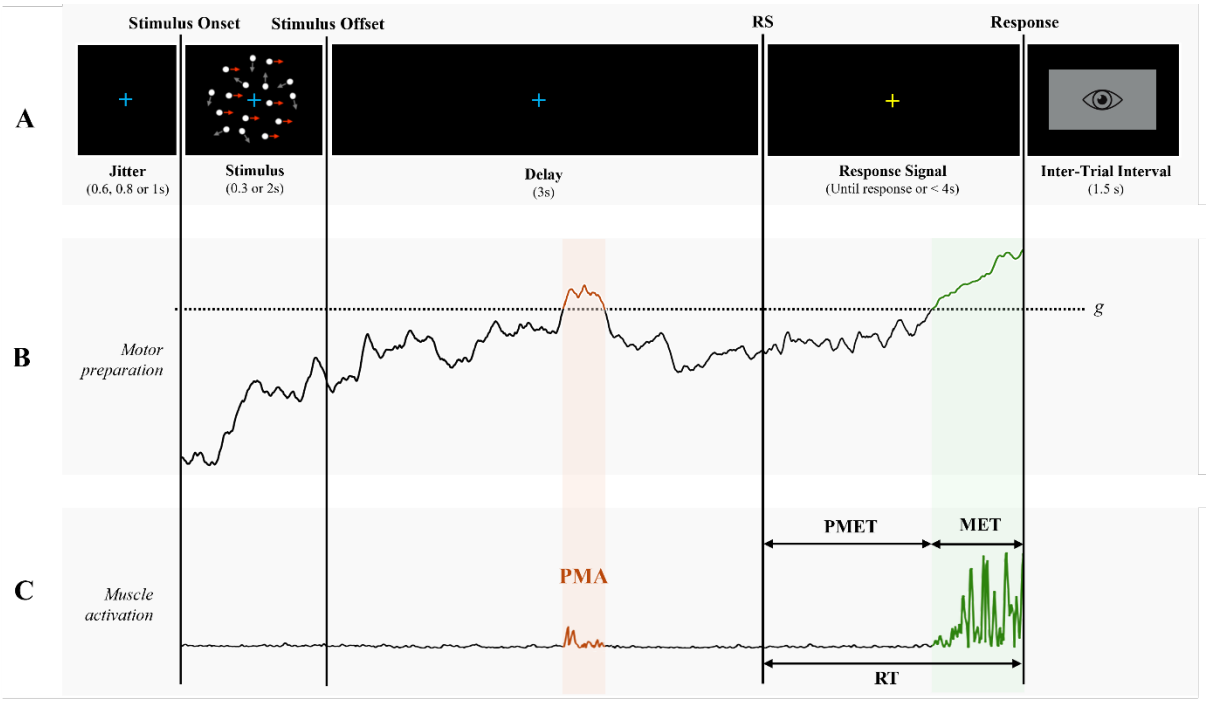


Fig 2. Experimental design and single-trial theoretical dynamics at the motor preparation and muscle activation levels. Panel A illustrates the time-course of a trial in our response signal (RS) variant of the random-dot motion task. See text for details. Single-trial dynamics predicted by the evidence accumulation theory at the levels of motor preparation and muscle activation are shown in Panels B and C, respectively. Trajectories represent one possible realization of the expected mean dynamics depicted in Fig. 1, reflecting the influence of moment-to-moment noise on individual trials. Muscle activation begins when motor preparation exceeds gating inhibition (parameter *g*). This activation can manifest as a partial muscle activation (PMA) when the drive is too weak to produce an overt response (orange shaded area), or as an overt response when the drive is strong enough (green shaded area). The response time (RT) corresponds to the latency between RS onset and the response. In the present study, it is decomposed into a pre-motor execution time (PMET) and a motor execution time (MET), based on the onset of muscle activation leading up to the response.

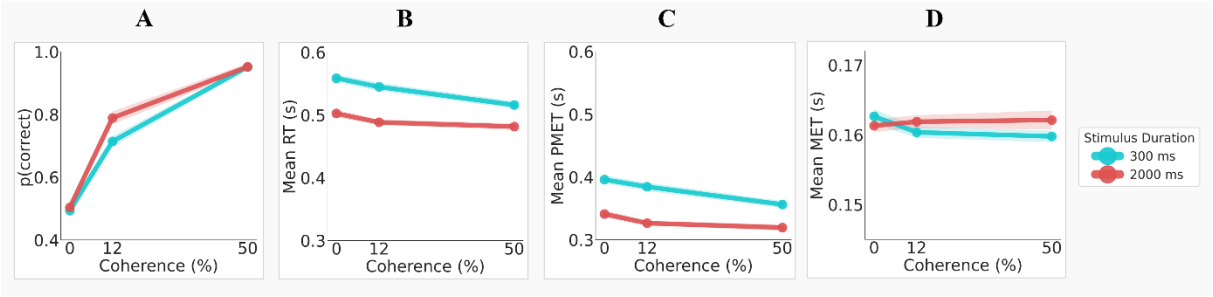


Fig 3. Behavioral outcomes and response time fractionation based on the onset of muscle activation preceding the response. Panels A-D show, respectively, response accuracy, mean response time (RT), mean pre-motor execution time (PMET), and mean motor execution time (MET) as a function of motion coherence and duration. Shaded areas represent ±1 within-subjects standard error of the mean, computed using the method proposed by Morey (2008). s = seconds.

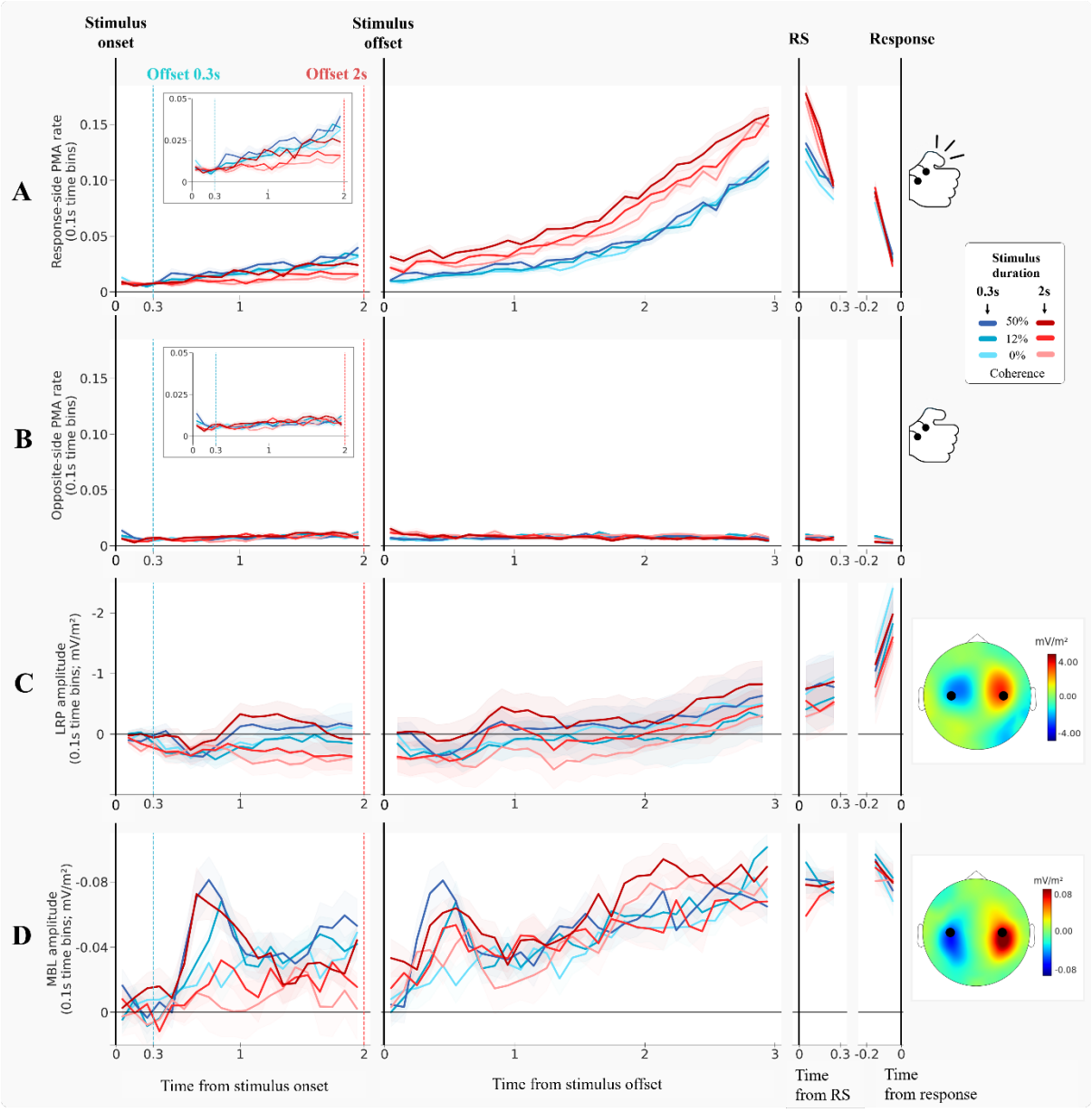


Fig 4. Rate of partial muscle activations and electroencephalographic measures of effector-selective motor preparation aligned to each trial event. A) Grand-average rate of partial muscle activations (PMAs) in the electromyographic channel of the responding effector as a function of motion coherence and duration. PMA rate was estimated as the mean number of PMAs per 0.1-s bin. From left to right, subplots show PMA rate aligned to stimulus onset, stimulus offset, RS onset, and the response, respectively. These locking events are indicated by the black vertical lines. The inset in the leftmost subplot provides a zoomed view of the corresponding data. The grand-average rate of PMAs in the opposite electromyographic channel is shown in panel B. C) Grand-average lateralized readiness potential (LRP) computed in 0.1 s bins, as a function of motion coherence and duration. The layout of the subplots matches that of panels A and B. The y-axis is inverted such that upward deflections indicate stronger effector-selective motor preparation. The map on the right shows the grand-average scalp topography in the -0.2 to 0 s interval relative to the response, collapsed across experimental conditions. It was computed by subtracting the mean current source density amplitude for right-hand trials from that for left-hand trials. Black circles indicate the electrode sites used to compute the LRP (C3 and C4). D) Grand-average mu–beta-band spectral amplitude lateralization

(MBL), computed in 0.1 s bins, as a function of motion coherence and duration. The y-axis is also inverted such that upward deflections indicate stronger effector-selective motor preparation. The layout of the subplots matches that of the other panels. The map on the right shows the grand-average scalp topography in the -0.2 to 0 s interval relative to the response, collapsed across experimental conditions. It was computed by subtracting the mean mu–beta-band spectral amplitude for right-hand trials from that for left-hand trials. For all panels, shaded areas represent ±1 within-subjects standard error of the mean, computed using the method proposed by Morey (2008). s = seconds; mV/m^2 = millivolts per square meter.

Tables

Table 1. Significant effects from the ANOVAs on response accuracy, mean response time (RT), mean pre-motor execution time (PMET), and mean motor execution time (MET).

Dependent variable	Significant effects	*p*-value
Accuracy	duration	$p < .001$
	coherence	$p < .001$
	duration × coherence	$p < .001$
Mean RT	duration	$p < .001$
	coherence	$p < .001$
	duration × coherence	$p = .016$
Mean PMET	duration	$p < .001$
	coherence	$p < .001$
	duration × coherence	$p = .035$
Mean MET	–	–

Note. Full ANOVA results, including effect sizes, are reported in Table S1.

Table 2. Significant effects from the ANOVAs on partial muscle activation (PMA) rate in the electromyographic channel of the responding effector, mu–beta-band spectral amplitude lateralization (MBL), and lateralized readiness potential (LRP) amplitude across task periods.

Dependent variable	Significant effects	Period			
		Stimulus	Delay	RS	Response
PMA rate	duration	$p = .001$	$p < .001$	$p < .001$	–
	coherence	$p = .021$	$p = .04$	–	–
	time	$p < .001$	$p < .001$	$p < .001$	$p < .001$
	duration × coherence	–	$p = .04$	–	–
	duration × time	$p < .001$	$p < .001$	$p = .015$	$p = .036$
	coherence × time	$p = .005$	_	_	_
MBL amplitude	time	$p < .001$	$p = .012$	–	$p < .001$
	coherence × time	$p < .001$	_	_	–
LRP amplitude	time	–	$p < .001$	$p = .001$	$p < .001$

Note. Entries indicate *p*-values for significant ANOVA effects. Full ANOVA results, including effect sizes, are reported in Tables S2–S4. Time refers to the time-bin factor. RS = response signal.

Supplemental Material

Neurocognitive Mechanisms Underlying Short-Term Delays in Perceptual Decision Reporting

Johan Achard, Thibault Gajdos Preuss, Mathieu Servant

Fig S1. Proportion of premature responses within a 0.4 s window centered on response signal onset as a function of motion coherence and duration, averaged across subjects. Error bars represent ±1 within-subjects standard error of the mean, computed using the method proposed by Morey (2008).

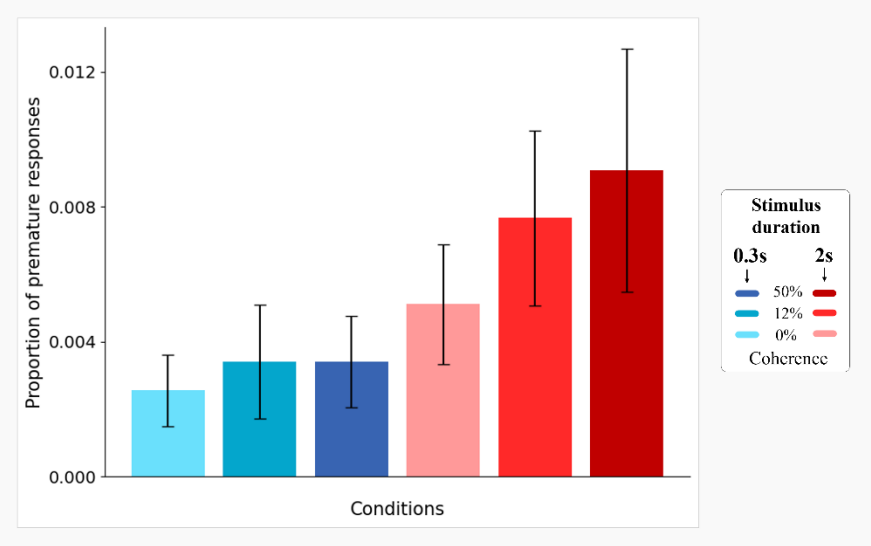


Table S1. Full ANOVA results for response accuracy, mean response time (RT), mean pre-motor execution time (PMET), and mean motor execution time (MET).

Dependent variable	Effect tested	df1	df2	*F*	*p*-value	η_p^2
Accuracy	duration	1	21	22.68	$p < .001$	0.519
	coherence	1.31	27.4	222.22	$p < .001$	0.914
	duration × coherence	2	42	12.17	$p < .001$	0.367
Mean RT	duration	1	21	65.79	$p < .001$	0.758
	coherence	1.5	31.39	14.19	$p < .001$	0.403
	duration × coherence	2	42	4.61	$p = .016$	0.180
Mean PMET	duration	1	21	69.70	$p < .001$	0.769
	coherence	2	42	14.11	$p < .001$	0.402
	duration × coherence	2	42	3.58	$p = .035$	0.147
Mean MET	duration	1	21	1.10	$p = .320$	–
	coherence	1.51	31.72	0.63	$p = .537$	–
	duration × coherence	2	42	2.44	$p = .104$	–

Note. A partial eta squared statistic (η_p^2) is reported as a measure of effect size. Degrees of freedom (df) with decimals indicate that a Greenhouse-Geisser correction was applied.

Table S2. Full ANOVA results for partial muscle activation rate in the electromyographic channel of the responding effector across task periods.

Period	Effect tested	df1	df2	*F*	*p*-value	η_p^2
Stimulus	duration	1	21	13.38	$p = .001$	0.389
	coherence	1.33	28.01	5.550	$p = .021$	0.200
	time	2.7	56.75	18.05	$p < .001$	0.462
	duration × coherence	2	42	0.312	$p = .734$	–
	duration × time	5.086	106.82	4.645	$p < .001$	0.181
	coherence × time	38	798	1.718	$p = .005$	0.076
	duration × coherence × time	38	798	1.058	$p = .377$	–
Delay	duration	1	21	33.19	$p < .001$	0.612
	coherence	2	42	3.36	$p = .044$	0.138
	time	1.21	25.43	31.42	$p < .001$	0.599
	duration × coherence	2	42	3.474	$p = .042$	0.142
	duration × time	5.38	112.986	4.520	$p < .001$	0.177
	coherence × time	38	798	0.737	$p = .879$	–
	duration × coherence × time	38	798	0.959	$p = .542$	–
Response signal	duration	1	21	36.56	$p < .001$	0.635
	coherence	1.56	32.82	1.159	$p = .315$	–
	time	1	21	21.96	$p < .001$	0.511
	duration × coherence	2	42	0.041	$p = .959$	–
	duration × time	1	21	7.046	$p = .015$	0.251
	coherence × time	2	42	0.407	$p = .668$	–
	duration × coherence × time	2	42	0.587	$p = .561$	–
Response	duration	1	21	0.011	$p = .919$	–
	coherence	2	42	0.724	$p = .491$	–
	time	1	21	38.31	$p < .001$	0.646
	duration × coherence	2	42	0.636	$p = .534$	–
	duration × time	1	21	5.013	$p = .036$	0.193
	coherence × time	2	42	1.172	$p = .320$	–
	duration × coherence × time	2	42	0.193	$p = .825$	–

Note. A partial eta squared statistic (η_p^2) is reported as a measure of effect size. Time refers to the time bin factor. Degrees of freedom (df) with decimals indicate that a Greenhouse-Geisser correction was applied.

Table S3. Full ANOVA results for the lateralized readiness potential amplitude across task periods.

Period	Effect tested	df1	df2	*F*	*p*-value	η_p^2
Stimulus	duration	1	21	0.636	$p = .434$	–
	coherence	2	42	1.524	$p = .230$	–
	time	4	83.98	1.876	$p = .122$	–
	duration × coherence	2	42	0.391	$p = .679$	–
	duration × time	3.06	64.21	1.181	$p = .141$	–
	coherence × time	38	798	1.177	$p = .217$	–
	duration × coherence × time	38	798	0.595	$p = .976$	–
Delay	duration	1	21	0.019	$p = .892$	–
	coherence	1.34	28.13	0.44	$p = .569$	–
	time	1.85	38.79	10.791	$p = .001$	0.339
	duration × coherence	2	42	0.378	$p = .687$	–
	duration × time	2.11	44.4	0.555	$p = .587$	–
	coherence × time	38	798	0.457	$p = .998$	–
	duration × coherence × time	38	798	0.408	$p = .999$	–
Response signal	duration	1	21	0.185	$p = .671$	–
	coherence	1.45	30.43	0.252	$p = .707$	–
	time	1	21	14.076	$p = .001$	0.401
	duration × coherence	2	42	0.267	$p = .767$	–
	duration × time	1	21	1.238	$p = .278$	–
	coherence × time	2	42	1.058	$p = .356$	–
	duration × coherence × time	2	42	0.108	$p = .898$	–
Response	duration	1	21	0.4	$p = .534$	–
	coherence	1.55	32.45	0.31	$p = .680$	–
	time	1	21	54.331	$p < .001$	0.721
	duration × coherence	2	42	0.666	$p = .519$	–
	duration × time	1	21	4.129	$p = .055$	–
	coherence × time	2	42	0.246	$p = .783$	–
	duration × coherence × time	2	42	0.589	$p = .560$	–

Note. A partial eta squared statistic (η_p^2) is reported as a measure of effect size. Time refers to the time bin factor. Degrees of freedom (df) with decimals indicate that a Greenhouse-Geisser correction was applied.

Table S4. Full ANOVA results for mu–beta-band spectral amplitude lateralization across task periods.

Period	Effect tested	df1	df2	*F*	*p*-value	η_p^2
Stimulus	duration	1	21	2.146	$p = .158$	–
	coherence	1.31	27.56	2.813	$p = .096$	–
	time	2.69	56.51	6.739	$p < .001$	0.243
	duration × coherence	2	42	0.755	$p = .476$	–
	duration × time	3.77	79.15	1.823	$p = .136$	–
	coherence × time	38	798	2.805	$p < .001$	0.118
	duration × coherence × time	38	798	0.81	$p = .787$	–
Delay	duration	1	21	3.177	$p = .089$	–
	coherence	1.29	27.12	1.257	$p = .285$	–
	time	1.59	33.4	5.642	$p = .012$	0.212
	duration × coherence	2	42	1.586	$p = .217$	–
	duration × time	6.07	127.39	1.078	$p = .379$	–
	coherence × time	38	798	1.008	$p = .458$	–
	duration × coherence × time	38	798	1.199	$p = .194$	–
Response signal	duration	1	21	3.039	$p = .096$	–
	coherence	2	42	0.16	$p = .853$	–
	time	1	21	0.056	$p = .810$	–
	duration × coherence	2	42	1.216	$p = .307$	–
	duration × time	1	21	1.509	$p = .233$	–
	coherence × time	2	42	0.024	$p = .977$	–
	duration × coherence × time	2	42	1.458	$p = .244$	–
Response	duration	1	21	0.121	$p = .731$	–
	coherence	2	42	0.207	$p = .814$	–
	time	1	21	16.547	$p < .001$	0.441
	duration × coherence	2	42	0.097	$p = .908$	–
	duration × time	1	21	3.126	$p = .092$	–
	coherence × time	2	42	0.584	$p = .562$	–
	duration × coherence × time	1.56	32.79	0.724	$p = .460$	–

Note. A partial eta squared statistic (η_p^2) is reported as a measure of effect size. Time refers to the time bin factor. Degrees of freedom (df) with decimals indicate that a Greenhouse-Geisser correction was applied.

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